Two-loop Type II amplitudes for $gg \to AA$ near threshold and the double parton scattering singularity

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In the limit of a heavy top quark, the two-loop amplitudes for $gg \to AA$ contain a class of diagrams, Type II, in which each pseudoscalar couples to the gluons through its own effective vertex. Their finite remainders are known in closed form, but their behaviour near the pair production threshold has not been studied. I expand them in the pseudoscalar velocity β_A at fixed scattering angle through β_A^6. Once the endpoint constants are expressed in a basis of eight transcendental constants, all inverse powers of β_A and the odd powers through β_A^5 cancel identically. At leading power the angle enters only through the logarithm of the transverse momentum p_T of the pseudoscalars. I identify this logarithm with the double parton scattering singularity of one-loop amplitudes: in the forward limit, and with the overall constant fixed at threshold, the velocity dependence of its one-loop coefficient follows the factorised formula of Gaunt and Stirling, including the helicity weights of gluon splitting and the sign structure of a CP-odd coupling, to 3×10^{-15}. An explicit one-loop calculation of $gg \to HH$ through two CP-even vertices gives the corresponding scalar weight with the same constant. At two loops the equal helicity amplitude grows as $\ln^3 p_T$, and the opposite helicity amplitude, which vanishes as β_A^2 at tree level and has no logarithm at one loop, as $\ln^2 p_T$. In the finite virtual corrections to the cross section the logarithm cancels at one loop; at two loops $\ln^3 p_T$ cancels and the highest power is $\ln^2 p_T$, with coefficient $-\frac{7}{4}\pi^2 N_c^2$. I also find that the published two-loop remainders are larger by a factor of 64 than the lower orders and that the published one-loop remainders omit their dependence on the renormalisation scale; all results are given in a consistent normalisation. The series reproduce the unexpanded remainders to 6.77×10^{-5} or better for $\beta_A \leq 0.2$ and are released with an evaluator.

I. INTRODUCTION

Precision tests of the Standard Model (SM) and searches for new particles at the Large Hadron Collider (LHC) require theoretical predictions for production rates and differential cross sections. The accuracy of these predictions is important when comparing different models or determining whether a discrepancy with a measurement indicates a new interaction. In gluon fusion processes, QCD radiative corrections can be sizeable, and calculations beyond leading order (LO) are needed to assess their effect on the predicted rates and distributions [1]. Loop amplitudes enter the virtual contributions to these corrections. Their calculation requires not only the evaluation of loop integrals, but also an understanding of their behaviour in the physical region. In particular, amplitudes can develop singularities at special kinematic configurations even when the corresponding tree amplitudes are regular. The location and strength of these singularities matter for numerical evaluation and for the behaviour of differential cross sections [2, 3]. An expansion about such a configuration allows the singular terms to be studied explicitly, together with the regular terms needed for a quantitative prediction.

A relevant example is neutral Higgs boson pair production in two-Higgs-doublet models (2HDM). At the LHC, the dominant production mechanism is gluon fusion through top quark loops, so that loop amplitudes enter already at LO [1, 4]. For pseudoscalar pairs, $gg \to AA$, the next-to-LO (NLO) QCD corrections were first computed in the limit of a heavy top quark [5], later matched to parton showers [6], and subsequently obtained with the full dependence on the top quark mass [1]. In the benchmark considered in Ref. [1], the full NLO cross section for AA lies about 5% below the Born-improved heavy top quark approximation at LHC energies, compared with about 12% for hH. For this benchmark, the comparison supports using the heavy top quark limit (HTL) as a starting point for higher orders, with the exact mass dependence retained at LO. Beyond NLO, only the two-loop virtual amplitudes are known in this limit [7]. For SM Higgs pairs, the corrections in the same limit are known to N^{3}LO [8–10]. The same effective gluon interactions also contribute to the production of axion-like particle pairs, and non-resonant $gg \to aa$ production has been used to constrain their interactions at the LHC [11].

In the limit of a heavy top quark, a pseudoscalar field $\Phi^A(x)$ couples to gluons and light quarks through the effective Lagrangian [12]

$$\mathcal{L}_{\text{eff}}^A = \Phi^A(x) \left[-\frac{1}{8} C_G \, O_G(x) - \frac{1}{2} C_J \, O_J(x) \right], \tag{1}$$

with the gluonic and light quark operators

$$O_G(x) = G^{a\mu\nu} \tilde{G}^a_{\mu\nu} = \epsilon_{\mu\nu\rho\sigma} G^{a\mu\nu} G^{a\rho\sigma}, \tag{2}$$

$$G^{a\mu\nu} = \partial^\mu G^{a\nu} - \partial^\nu G^{a\mu} + g_s f^{abc} G^\mu_b G^\nu_c, \tag{3}$$

where f^{abc} are the structure constants of SU(3) and $\epsilon_{\mu\nu\rho\sigma}$ is the Levi-Civita tensor. The light quark operator is the divergence of the flavour singlet axial vector current,

$$O_J(x) = \partial_\mu \left(\bar{\psi} \gamma^\mu \gamma^5 \psi \right). \tag{4}$$

By the Adler–Bardeen theorem [13], C_G receives no QCD corrections beyond one loop, whereas C_J starts at second order in the strong coupling $a_s \equiv g_s^2/16\pi^2 = \alpha_s/4\pi$. The Wilson

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coefficients are

$$C_G\left(a_s\right) = -a_s 2^{\frac{5}{4}} G_F^{\frac{1}{2}} \cot\beta\,, \tag{5}$$

$$C_J\left(a_s\right) = -\left[a_s C_F\left(\frac{3}{2} - 3\ln\frac{\mu_R^2}{m_t^2}\right) + a_s^2 C_J^{(2)} + ...\right] C_G\,, \tag{6}$$

where G_F is the Fermi constant, m_t the top quark mass and $\tan\beta$ the ratio of the vacuum expectation values of the two Higgs doublets, in a model without spontaneous CP violation; this angle is unrelated to the velocity β_A of Eq. (8). C_F is the quadratic Casimir of the fundamental representation and μ_R the scale at which a_s is renormalised. The conventions are those of Ref. [7]. Operator mixing, ultraviolet renormalisation and the treatment of γ_5 [14] follow the original calculation in Ref. [7]; the same effective theory underlies inclusive pseudoscalar production at NNLO [15].

I consider the dominant gluon fusion subprocess

$$g(p_1) + g(p_2) \to A(p_3) + A(p_4), \tag{7}$$

with incoming gluon momenta $p_{1,2}^2 = 0$ and outgoing pseudoscalar momenta $p_{3,4}^2 = m_A^2$. The amplitude for $gg \to AA$ separates into two classes [7]. In Type I both pseudoscalars originate from a single effective vertex. In Type II each pseudoscalar is emitted from its own vertex and the two vertices are connected by gluon exchange in the t or u channel (Fig. 1a). In the full theory the Type II diagrams correspond to two top quark triangles joined by a gluon, the one-particle reducible contributions of Refs. [1, 5]; for Higgs pairs the corresponding class was computed at two loops in Ref. [9]. The two-loop Type II finite remainders for $gg \to AA$ were obtained in closed form in Ref. [7]. They contain polylogarithms of weight up to four, among them $\mathrm{Li}_{2,2}$, and inverse powers of the Gram determinant $tu - m_A^4$, which vanishes at the pair threshold. The expressions occupy about 204 MB, and large cancellations between their terms make them hard to evaluate close to threshold.

There are two reasons to study this challenging threshold region. The first is phenomenological. In the NLO predictions of Ref. [1] for $m_A = 134.7$ GeV the invariant mass distribution of the pair is larger at $m_{AA} = 300$ GeV, about 30 GeV above threshold, than at 400 GeV, and threshold resummation of this distribution requires at each m_{AA} a hard function obtained from the infrared (IR) finite virtual amplitude [16–20]. The second concerns the analytic structure of loop amplitudes. When the transverse momentum p_T of the final state particles vanishes, a one-loop amplitude with massless internal lines can have a Landau singularity [21] in which each incoming parton splits into two collinear partons that then take part in two separate $2 \to 1$ scatterings. Gaunt and Stirling showed that this double parton scattering (DPS) singularity is at most logarithmic in p_T for SM amplitudes and that, for the processes they considered, it occurs only when the incoming state has $J_z = 0$. They also derived a factorised formula for its coefficient [2]. Dixon and Esterlis studied the corresponding singularity of six-gluon amplitudes in planar $\mathcal{N} = 4$ super Yang–Mills theory to high loop order [3]. As far as I am aware, it has not been examined beyond one loop in QCD at the level of amplitudes for a $2 \to 2$ process with a colour singlet final state; NLO corrections to the underlying splitting mechanism are known for double parton distributions [22]. The Type II amplitude for $gg \to AA$ contains diagrams with this topology: each incoming gluon can split into two gluons, which fuse pairwise into the two pseudoscalars at the effective vertices (Fig. 1b). At threshold, $p_T \to 0$ at every scattering angle, so a threshold expansion probes this configuration directly.

In hadronic collisions, a configuration in which a parton from each proton splits into two can enter both the single and double parton scattering calculations. The overlap must therefore be subtracted to avoid counting the same contribution twice [22, 23]. DPS also receives a contribution in which only one of the two parton pairs arises from a perturbative splitting, while the other proton supplies two partons without such a splitting. This contribution has been studied for Higgs boson pairs [24]. DPS has also been observed in same-sign W pair production at the LHC [25, 26]. The overlap between single and double parton scattering has been re-examined using two-loop splitting kernels [27].

In this paper, I expand the two-loop Type II finite remainders of Ref. [7] in the velocity

$$\beta_A = \sqrt{1 - 4m_A^2/\hat{s}} \tag{8}$$

of the pseudoscalars in the partonic centre of mass frame, at fixed scattering angle, through β_A^6. The main results are the following.

1. After the constants that arise in the expansion are expressed in a basis of eight transcendental constants, all inverse powers of β_A and the odd powers through β_A^5 cancel, and every coefficient is even in $\cos\theta$, as required by Bose symmetry. At leading power the angle enters only through the logarithm of the transverse momentum p_T of the pseudoscalars.

2. This logarithm is the DPS singularity. At one loop its coefficient follows the factorised formula of Ref. [2] for gluon splitting and the opposite signs of the amplitudes for $g^+g^+ \to A$ and $g^-g^- \to A$: with the overall constant fixed at threshold, it reproduces the velocity dependence of the exact one-loop coefficient in the forward limit. A one-loop calculation of $gg \to HH$ confirms the corresponding weight for a CP-even coupling.

3. At two loops the $J_z = 0$ amplitude, with equal gluon helicities, grows as $\ln^3 p_T$, and the ratio of the leading logarithms at two and one loops is the one found to all orders in planar $\mathcal{N} = 4$ super Yang–Mills theory [3]. The $|J_z| = 2$ amplitude, with opposite helicities, has no DPS logarithm at one loop but grows as $\ln^2 p_T$ at two loops; I leave the origin of this logarithm open.

4. In the finite virtual corrections to the cross section the one-loop logarithm cancels. At two loops $\ln^3 p_T$ cancels, and the highest power is $\ln^2 p_T$.

All results are given in a consistent normalisation. The series reproduce the unexpanded remainders with relative errors of at

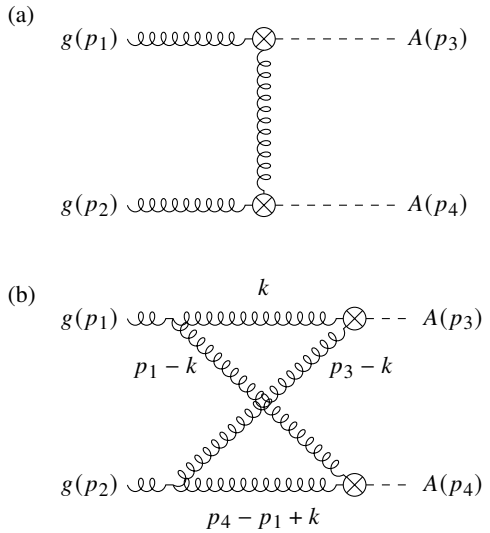


FIG. 1. (a) Type II tree diagram in the t channel; the u-channel diagram is obtained by exchanging the pseudoscalars. The crossed circles are the effective vertices of O_G. (b) One-loop Type II diagram with loop momentum k. Its double parton scattering singularity lies at $k = xp_1$ and $p_3 - k = (1-x)p_2$, where the four internal gluons are on shell and collinear with the incoming gluons: each incoming gluon splits into two, which fuse pairwise at the effective vertices. Gaunt and Stirling call this topology the crossed box [2].

most 6.77×10^{-5} for $\beta_A \leq 0.2$; they are released together with an evaluator [28].

The paper is organised as follows. Section II sets out the conventions, the normalisation of the remainders and the counting of orders. Section III summarises the procedure and describes the expansion and its accuracy. Section IV discusses the leading power terms and their relation to the DPS singularity, and Sec. V their consequences for the cross section. The conclusions are in Sec. VI.

II. AMPLITUDES AND CONVENTIONS

A. Theoretical framework

To study the threshold behaviour of $gg \to AA$ through two loops, I use the amplitudes calculated in Ref. [7] with the effective Lagrangian of Eq. (1). The labels Type I and Type II follow that reference and distinguish the two classes of diagrams. They are unrelated to the Yukawa types of two-Higgs-doublet models. The Type II tree diagrams correspond, in the full theory, to two top quark triangles connected by a gluon, i.e. the one-particle reducible contribution of Ref. [1]. Here I consider the contribution with two O_G insertions and omit the superscript II from the amplitudes. Diagrams with one O_G and one O_J insertion first contribute at one loop. Their coefficients are proportional to the O_G–O_G tree coefficients, as discussed in Sec. II D.

B. Kinematics

The Mandelstam variables are

$$\hat{s} = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_2 - p_3)^2, \tag{9}$$

with $\hat{s} + t + u = 2m_A^2$. Following the conventions of Ref. [7], I use

$$\hat{s} = m_A^2 \frac{(1+x)^2}{x}, \qquad t = -m_A^2 y, \qquad u = -m_A^2 z, \tag{10}$$

with $x + x^{-1} = y + z$. With $c = \cos\theta$ in the partonic centre of mass frame,

$$\hat{s} = \frac{4m_A^2}{1-\beta_A^2}, \qquad t, u = m_A^2 - \frac{\hat{s}}{2}(1 \mp \beta_A c), \tag{11}$$

and the threshold substitution is

$$x = \frac{1-\beta_A}{1+\beta_A}, \qquad y, z = \frac{1+\beta_A^2 \mp 2\beta_A c}{1-\beta_A^2}. \tag{12}$$

All three variables tend to one as $\beta_A \to 0^+$ at fixed c, and the expansion has to follow this correlated limit. The Gram determinant and the transverse momentum are

$$tu - m_A^4 = \frac{4m_A^4\beta_A^2(1-c^2)}{(1-\beta_A^2)^2}, \qquad p_T^2 = \frac{m_A^2\beta_A^2(1-c^2)}{1-\beta_A^2}. \tag{13}$$

Both vanish at threshold at every angle, and also at fixed β_A when $c \to \pm 1$.

C. Tensor decomposition and helicity amplitudes

With the overall factor C_G^2 removed, the amplitude is decomposed as

$$\mathcal{M}_{ab}^{\mu\nu} = \delta_{ab}\left(T_1^{\mu\nu}\mathcal{M}_1 + T_2^{\mu\nu}\mathcal{M}_2\right), \tag{14}$$

with

$$T_1^{\mu\nu} = g^{\mu\nu} - \frac{p_1^\nu p_2^\mu}{p_1\cdot p_2}, \tag{15}$$

$$T_2^{\mu\nu} = g^{\mu\nu} + \frac{1}{(p_1\cdot p_2)p_T^2}\Big[m_A^2 p_2^\mu p_1^\nu - 2(p_1\cdot p_3)p_2^\mu p_3^\nu - 2(p_2\cdot p_3)p_3^\mu p_1^\nu + 2(p_1\cdot p_2)p_3^\mu p_3^\nu\Big], \tag{16}$$

$$p_T^2 = \frac{tu - m_A^4}{\hat{s}}. \tag{17}$$

For physical gluon polarisations, T_1 contributes only to equal helicities ($J_z = 0$) and T_2 only to opposite helicities ($|J_z| = 2$), each with a contraction of unit modulus. Up to phases the helicity amplitudes are [29]

$$\mathcal{A}_{\pm\pm} = C_G^2\,\mathcal{M}_1, \qquad \mathcal{A}_{\pm\mp} = C_G^2\,\mathcal{M}_2\,e^{\pm 2i\phi}, \tag{18}$$

where ϕ is the azimuth of $\vec{p}_3$, and $\sum_{\text{hel,col}} |\mathcal{A}|^2 = 2(N_c^2 - 1)\, C_G^4\, (|\mathcal{M}_1|^2 + |\mathcal{M}_2|^2)$. The labels 1 and 2 refer to the tensors, not to the diagram classes.

At tree level the two Type II diagrams give

$$\begin{aligned} \mathcal{M}_1^{(0)} &= \frac{i\hat{s}}{2}, \\ \mathcal{M}_2^{(0)} &= \frac{i m_A^2 (x-z)(x^2+1)(xz-1)}{4xz(x^2-xz+1)} \\ &= 2i m_A^2 \beta_A^2 (1-c^2) + O(\beta_A^4), \end{aligned} \tag{19}$$

in the variables of Eq. (10). I obtained Eq. (19) from the projections of the original calculation and checked it against a direct evaluation of the two Feynman diagrams with the vertex $iC_G \varepsilon_{\mu\alpha\rho\beta} k_1^\mu k_2^\rho \delta^{ab}$ that follows from Eq. (1) (App. B). The suppression of $\mathcal{M}_2^{(0)}$ by β_A^2 is the centrifugal barrier of the $|J_z| = 2$ state.

D. Finite remainders and normalisation

The coefficients $\mathcal{M}_i$ of Eq. (18), from which C_G^2 has been factored out, are expanded as

$$\mathcal{M}_i = \mathcal{M}_i^{(0)} + a_s \mathcal{M}_i^{(1)} + a_s^2 \mathcal{M}_i^{(2)} + \cdots, \tag{20}$$

with a_s renormalised at μ_R. The finite remainders, called the finite part in Ref. [7], are defined by the IR subtraction of Refs. [7, 30],

$$\begin{aligned} \mathcal{M}^{(1)} &= 2I^{(1)} \mathcal{M}^{(0)} + \mathcal{M}^{(1),\text{fin}}, \\ \mathcal{M}^{(2)} &= 4I^{(2)} \mathcal{M}^{(0)} + 2I^{(1)} \mathcal{M}^{(1)} + \mathcal{M}^{(2),\text{fin}}. \end{aligned} \tag{21}$$

From here on $\mathcal{M}^{(n)}$ denotes the finite remainder $\mathcal{M}^{(n),\text{fin}}$.

At two loops, I decompose each finite remainder into powers of the number of light quark flavours,

$$\mathcal{M}_i^{(2)} = \mathcal{M}_{i,0}^{(2)} + n_f\, \mathcal{M}_{i,1}^{(2)} + n_f^2\, \mathcal{M}_{i,2}^{(2)}. \tag{22}$$

Each $\mathcal{M}_{i,k}^{(2)}$ is finite, but none is the amplitude of a separate process: the term linear in n_f, for example, combines the diagrams with a closed quark loop with the terms of the renormalisation and of the infrared subtraction that depend on n_f. In the rest of the paper, the complete coefficient means $\mathcal{M}_i^{(2)}$ at $n_f = 5$. Furthermore, the remainders released with Ref. [7] are normalised differently at different orders: the two-loop remainders carry an overall factor 64 relative to the tree and one-loop amplitudes, and the one-loop remainders are those for $\mu_R = 1$ GeV. In the normalisation of Eq. (20) the remainders are therefore

$$\begin{aligned} \mathcal{M}_i^{(1)} &= \mathcal{M}_{i,\text{publ}}^{(1)} + 2\beta_0 \ln \frac{\mu_R^2}{\text{GeV}^2}\, \mathcal{M}_i^{(0)}, \\ \mathcal{M}_i^{(2)} &= \frac{1}{64}\, \mathcal{M}_{i,\text{publ}}^{(2)}, \end{aligned} \tag{23}$$

with $\beta_0 = (11N_c - 2n_f)/3$ and masses in GeV in the published expressions. The one-loop term restores the scale dependence required by renormalisation group invariance; with it, the released one and two-loop remainders are consistent with each other at every tested point. Appendix B gives the evidence. The logarithmic structure studied below, the expansion and its accuracy do not depend on these corrections; the one-loop constant and the finite virtual corrections of Sec. V do.

The one-loop amplitude with one O_G and one O_J insertion enters at $O(a_s^4)$, the order of the two-loop O_G–O_G amplitude, since $C_J = O(a_s)\, C_G$. It is proportional to the tree amplitude: in the normalisation of Eqs. (3.6) and (3.7) of Ref. [7],

$$\mathcal{M}_{i,GJ,\text{publ}}^{(1),\text{fin}} = -48\, n_f\, \mathcal{M}_i^{(0)}, \qquad i = 1, 2, \tag{24}$$

at every point of the physical phase space. The proportionality is consistent with the axial anomaly, which relates the divergence of the flavour singlet axial current to the gluonic pseudoscalar operator [13, 14, 31]. At $O(a_s^4)$ the mixed contribution therefore has the kinematic dependence of the tree amplitude and adds no logarithm of p_T.

E. Perturbative order

The expansion in Eq. (20) is written for the coefficients $\mathcal{M}_i$, with the overall factor C_G^2 removed. The superscript (n) counts loops of gluons and light quarks in the effective theory; in a two-Higgs-doublet model the top quark loops are contained in the Wilson coefficients. The perturbative order of a contribution to the cross section also depends on the powers of a_s in the Wilson coefficients.

Since $C_G \propto a_s$, the Type II tree, one-loop and two-loop contributions to the full amplitude are of order a_s^2, a_s^3 and a_s^4, respectively. In a 2HDM, the leading amplitude is of Type I and starts at $O(a_s)$, giving a LO cross section of order a_s^2. Interference between the leading Type I amplitude and the Type II tree amplitude contributes at order a_s^3, that is at NLO. This is the one-particle reducible contribution to the virtual corrections in Refs. [1, 5]. The Type II two-loop amplitude first enters through its interference with the leading Type I amplitude at order a_s^5, or N^{3}LO. The same counting applies to the corresponding class of amplitudes for Higgs pairs [9].

The mixed O_G–O_J contribution discussed in Sec. II D also contributes at this N^{3}LO. In the heavy top quark theory, $C_J = O(a_s)\, C_G$ [7], so its one-loop amplitude is of order a_s^4, the same as the two-loop O_G–O_G amplitude.

When no Type I amplitude is present, the Type II tree amplitude instead gives the LO cross section. This is the case for an axion-like particle whose only relevant interaction is its coupling to gluons through $G\tilde{G}$. The one-loop and two-loop terms in Eq. (20) then provide virtual contributions at NLO and NNLO, respectively. The present results therefore supply the O_G–O_G part of the two-loop virtual correction at NNLO for this process. A contact interaction of dimension six containing two pseudoscalars and two gluon fields would introduce an additional Type I amplitude, as in Ref. [11].

F. Finite top quark mass effects

The threshold expansion and the heavy top quark approximation involve different limits. Taking $\beta_A \to 0$ places the pair invariant mass close to $2m_A$, but imposes no restriction on m_A/m_t. Finite top quark mass effects must therefore be assessed separately when A is the pseudoscalar of a 2HDM, at values of $\tan\beta$ for which the top quark loop dominates. No heavy top quark approximation is involved when the coupling to $G\tilde{G}$ is specified directly as an effective interaction, as in the axion-like particle case above.

For the Type II tree amplitude, the dependence on the top quark mass is known. Each triangle in the full theory couples a pseudoscalar to one on-shell gluon and one virtual gluon. The two triangles in a given channel have the same form factor, so the contribution of each channel is multiplied, relative to its heavy top quark limit, by

$$\left[\frac{I_2(\tau_A,\lambda)}{I_2^\infty}\right]^2, \quad \tau_A = \frac{4m_t^2}{m_A^2}, \quad \lambda = \frac{4m_t^2}{t} \quad \text{or} \quad \frac{4m_t^2}{u}. \tag{25}$$

Here I_2 is the loop function that also appears in $A \to Z\gamma$, given in Ref. [1], and I_2^∞ denotes its heavy top quark limit. At the pair threshold, $t = u = -m_A^2$, so both channels have the same factor: 1.006, 1.013 and 1.022 for $m_A = 125$, 150 and 170 GeV, respectively, with $m_t = 172.5$ GeV. These differ from unity by at most about 2.2%.

For the loop corrections, a partial estimate can be made for the one-loop DPS logarithm. Its coefficient involves the amplitudes for $gg \to A$ with both gluons on shell, as discussed in Sec. IV C. The relevant triangle form factor is therefore obtained in the limit $\lambda \to \infty$, rather than at the spacelike virtuality of the exchanged gluon in the tree diagrams:

$$\frac{I_2(\tau_A,\infty)}{I_2^\infty} = \tau_A f(\tau_A), \qquad f(\tau) = \arcsin^2\!\left(\frac{1}{\sqrt{\tau}}\right). \tag{26}$$

If the DPS factorisation applies to the full theory, with the top quark triangles treated as hard subgraphs, the coefficient of the one-loop logarithm is multiplied by $[\tau_A f(\tau_A)]^2$. This gives 1.096, 1.145 and 1.196 for the same three masses, while the velocity dependence and the helicity selection rule would remain unchanged. Neither this estimate nor the comparison of tree amplitudes determines the finite top quark mass corrections to the remaining one-loop terms or to the complete two-loop amplitude.

III. THRESHOLD EXPANSION

The calculation has four steps, each checked independently. First, the finite remainders of Ref. [7] are split into their flavour coefficients and brought to the normalisation of Eq. (23) (App. B). Second, each flavour coefficient is expanded about $\beta_A = 0$ at fixed angle, through β_A^4 and, independently, through β_A^6 (Sec. III A). Third, the constants that arise at the endpoints are reduced to a basis of eight transcendental constants, after which the inverse and odd powers of β_A cancel identically (Sec. III B and App. A). Fourth, the series are compared with the unexpanded remainders, evaluated in Mathematica at increasing working precision and, at selected points, with an independent Python code (Sec. III C). The leading power terms obtained in this way are analysed in Sec. IV, where the forward limit of the one-loop remainders and a one-loop calculation of $gg \to HH$ test their relation to the DPS singularity, and their consequences for the cross section in Sec. V. The software and the computing environment are described in App. D.

A. Construction

The remainders of Ref. [7] are functions of x, y and z, Eq. (10). I replace these variables by their expressions in β_A and c from Eq. (12), and expand each flavour coefficient about $\beta_A = 0$ at fixed c,

$$\mathcal{M}_{i,r}^{(2)}(\beta_A, c) = \sum_{n=-4}^{p} \sum_{k} C_{nk}^{(i,r)}\, \beta_A^n \ln^k \beta_A + O_{\log}(\beta_A^{p+1}), \tag{27}$$

with $r = 0, 1, 2$ and $p = 4$ or 6. The coefficients $C_{nk}^{(i,r)}$ depend on c, m_A^2, μ_R^2 and N_c, and $O_{\log}(\beta_A^n)$ stands for β_A^n times a polynomial in $\ln\beta_A$. Inverse powers appear in intermediate steps because the tensor basis and the rational prefactors contain inverse Gram determinants, Eq. (13). They must be kept until all contributions at a given power have been combined.

The remainders contain logarithms, classical polylogarithms and the function $\mathrm{Li}_{2,2}$ of weight four [32, 33], whose arguments approach correlated endpoints as $\beta_A \to 0$. I expand the rational prefactors exactly and obtain the expansions of the special functions from differential equations in β_A, with boundary values at $\beta_A = 0$. These equations have the iterated-integral form familiar from harmonic polylogarithms [34] and from canonical differential equations for Feynman integrals [35]; the map $x = (1-\beta_A)/(1+\beta_A)$ is one of the transformations under which harmonic polylogarithms are closed [34]. The method thus expands the known remainders, not the loop integrals, which could instead be expanded near threshold by the method of regions [36]. For the dilogarithm, for example,

$$\mathrm{d}\,\mathrm{Li}_2(x)/\mathrm{d}\beta_A = 2(1-\beta_A^2)^{-1} \ln[2\beta_A/(1+\beta_A)] \text{ with} \tag{28}$$

$$\mathrm{Li}_2(1) = \pi^2/6 \text{ gives} \tag{29}$$

$$\mathrm{Li}_2(x) = \frac{\pi^2}{6} + 2\beta_A\big[\ln(2\beta_A) - 1\big] - \beta_A^2 + \beta_A^3\left[\frac{2}{3}\ln(2\beta_A) + \frac{1}{9}\right] - \frac{2}{3}\beta_A^4 + O\Big(\beta_A^5 \ln\beta_A\Big). \tag{30}$$

Since the prefactors start at β_A^{-4}, the special functions have to be expanded four orders beyond the final accuracy. The fourth and sixth order expansions were constructed independently, and all coefficients common to both agree.

B. Endpoint constants and exact cancellations

The boundary values at $\beta_A = 0$ introduce thirteen constants $\mathrm{MB}[p, q, a, w]$, defined in Eq. (A1) of App. A. Their arguments take the values ± 1 and $\frac{1}{2}$, so they are alternating Euler sums of weight three and four [37]. An integer relation search with the PSLQ algorithm [38] at 110 digits expresses each of them in the basis

$$\{\zeta_3,\ \pi^2 \ln 2,\ \ln^3 2,\ \pi^4,\ \pi^2 \ln^2 2,\ \ln^4 2,\ \mathrm{Li}_4(\tfrac{1}{2}),\ \zeta_3 \ln 2\}, \quad (31)$$

with a largest residual of 4×10^{-107} (Table VII).

With these reductions, and with the dilogarithms of $(1 - c)/2$, $(c - 1)/(c + 1)$ and $(c + 1)/(c - 1)$ rewritten through $\mathrm{Li}_2\big((1 + c)/2\big)$ by the reflection and Landen identities, every coefficient $C_{nk}^{(i,r)}$ becomes a polynomial in

$$\Big\{\ln \beta_A,\ \ln(1 \pm c),\ \mathrm{Li}_2\big(\tfrac{1+c}{2}\big),\ \ln \tfrac{m_A^2}{\mu_R^2}\Big\} \cup \{\ln 2,\ \pi,\ \zeta_3,\ \mathrm{Li}_4(\tfrac{1}{2})\} \quad (32)$$

with coefficients rational in c and N_c. Treating these quantities as algebraically independent, I find that

1. the coefficients of $\beta_A^{-4}, \ldots, \beta_A^{-1}$ vanish identically in every flavour coefficient;
2. the coefficients of β_A, β_A^3 and β_A^5 vanish identically, so that through β_A^6 the expansion is a series in β_A^2 with logarithms;
3. the coefficients of β_A^0, β_A^2, β_A^4 and β_A^6 of all six flavour coefficients, 24 in total, are even in c, so that Bose symmetry holds order by order.

The cancellations and the reductions of the constants were confirmed independently in Mathematica.

The cancellations have a simple origin. Exchanging the pseudoscalars exchanges t and u, and the invariants that are symmetric under this exchange,

$$\hat{s} = \frac{4m_A^2}{1 - \beta_A^2}, \qquad (t - u)^2 = \hat{s}^2 \beta_A^2 c^2, \qquad tu - m_A^4 = \hat{s}\, p_T^2, \quad (33)$$

with $p_T^2 = \hat{s}\beta_A^2(1 - c^2)/4$, depend on β_A only through β_A^2. The loops contain no massive particles, so there is no threshold singularity at $\hat{s} = 4m_A^2$, and the colourless final state has no Coulomb singularity. Near threshold, the only singularity is then the one at $p_T \to 0$ discussed in Sec. IV C, which is logarithmic at one loop [2]. The Gram determinant $tu - m_A^4$ vanishes at the same point, but its inverse powers come from the tensor basis and the rational prefactors and are not singularities of the amplitude. A remainder of the form $\sum_k a_k \ln^k p_T^2$, with coefficients a_k that are analytic in the invariants of Eq. (33) at threshold, is a series in β_A^2 with logarithms and has no inverse powers. This explains the form of the cancellations found above, but at two loops it is not an independent proof, since there the logarithmic nature of the singularity is established only by the calculation itself. Table I lists the leading terms. Since the odd powers vanish, the first omitted term is expected to be $O_{\log}(\beta_A^{p+2})$ rather than the $O_{\log}(\beta_A^{p+1})$ of Eq. (27), so the truncation errors should scale as β_A^6 and β_A^8 for $p = 4$ and $p = 6$, respectively, up to logarithmic factors.

TABLE I. Leading terms of the threshold expansion of the two-loop Type II remainders. The second column lists the inverse powers with nonzero coefficients in the construction files; they cancel identically after the reductions of Sec. III B, as do all odd powers. The last column gives the powers of $\mathcal{L}$ at leading power.

quantity	inverse powers in construction	leading power	powers of $\mathcal{L}$
$\mathcal{M}_{1,0}^{(2)}$	$\beta_A^{-2}, \beta_A^{-1}$	β_A^0	0, 1, 2, 3
$\mathcal{M}_{1,1}^{(2)}$	–	β_A^0	0, 1, 2
$\mathcal{M}_{1,2}^{(2)}$	–	β_A^0	0
$\mathcal{M}_{2,0}^{(2)}$	$\beta_A^{-4}, \beta_A^{-3}, \beta_A^{-2}, \beta_A^{-1}$	β_A^0	0, 1, 2
$\mathcal{M}_{2,1}^{(2)}$	β_A^{-1}	β_A^0	0, 1
$\mathcal{M}_{2,2}^{(2)}$	–	β_A^2	0

The coefficient of n_f^2 is the simplest:

$$\mathcal{M}_{1,2}^{(2)}(\beta_A, c) = K + O(\beta_A^2), \quad (34)$$

$$\mathcal{M}_{2,2}^{(2)}(\beta_A, c) = \beta_A^2(1 - c^2)K + O(\beta_A^4), \quad (35)$$

where K does not depend on the angle. It is the only flavour coefficient that follows the tree-level pattern $\mathcal{M}_2 \propto \beta_A^2(1 - c^2)$ of Eq. (19), with the same ratio to the tree amplitude in both tensor structures. The complete expressions through β_A^6 are released as Wolfram Language files (App. D).

C. Accuracy

I compared the series with the unexpanded remainders on the grid in Table II, which contains 9 values of β_A between 10^{-3} and $3/10$, 5 scattering angles and 3 combinations of mass, renormalisation scale and number of colours. At each of the 135 points, the comparison included the six flavour coefficients and the two complete coefficients, giving 1080 comparisons in total. Each unexpanded remainder was evaluated with 80, 120, 160, 220 and 300 digits of working precision, and a value was accepted once two successive precisions agreed to a relative difference below 10^{-25}, with at least 30 significant digits in both. The parameter set with $N_c \neq 3$ was included to test the colour dependence; all physical results use $N_c = 3$. The relative truncation error is defined as

$$E_i^{(p)} = \frac{\left|\mathcal{M}_i^{[p]} - \mathcal{M}_i\right|}{|\mathcal{M}_i|}, \qquad p = 4, 6, \quad (36)$$

where $\mathcal{M}_i$ is the unexpanded remainder and $\mathcal{M}_i^{[p]}$ its series through β_A^p. Figure 2 shows these errors for the complete coefficients at $n_f = 5$, $\mathcal{M}_1$ in the left column and $\mathcal{M}_2$ in the right. At each β_A the maximum over the angles and parameter sets of Table II is plotted, with circles for the series through β_A^4 and squares for the series through β_A^6. For the series through β_A^6

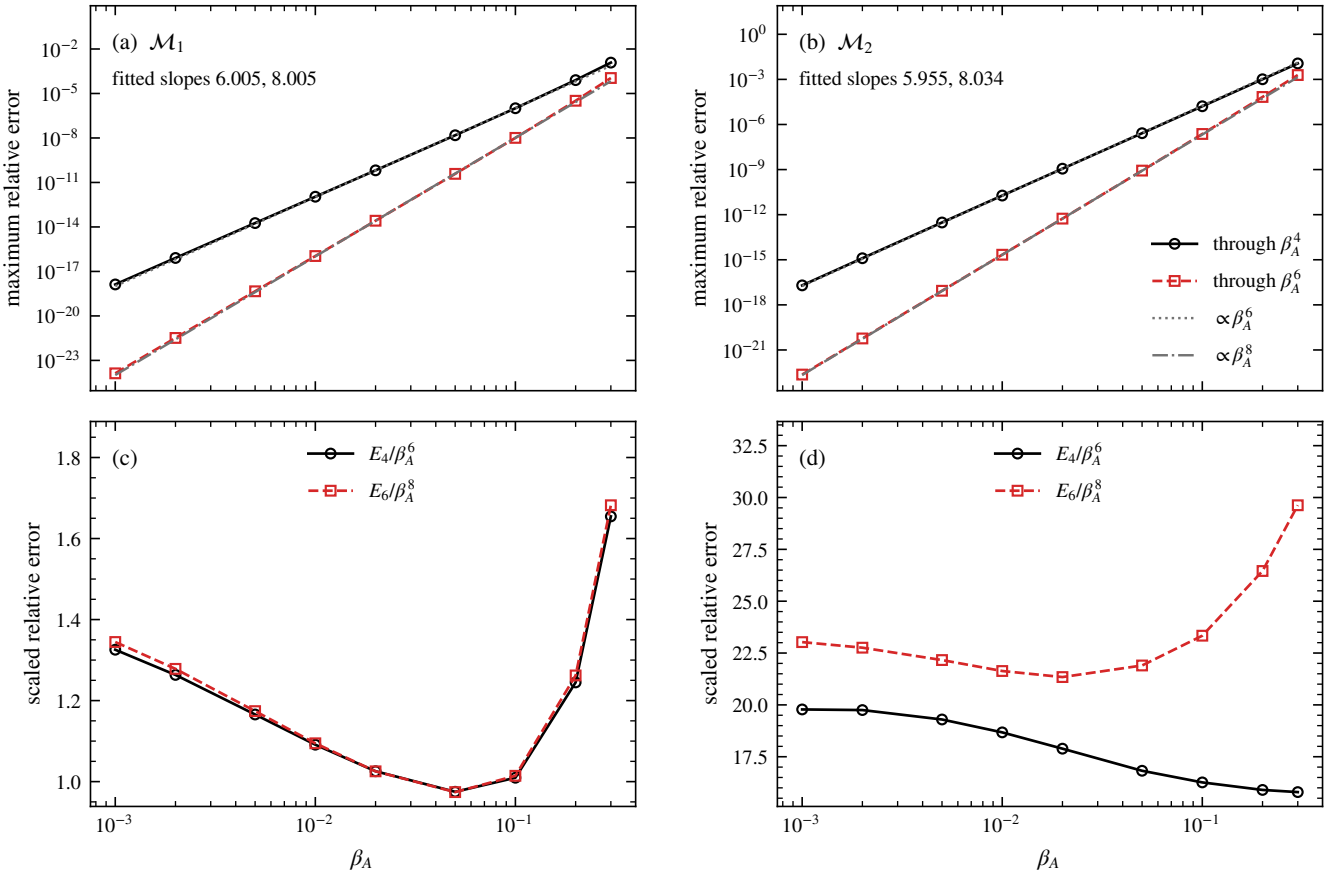


FIG. 2. Relative truncation errors, Eq. (36), of the complete two-loop coefficients at $n_f = 5$: $\mathcal{M}_1$ (left) and $\mathcal{M}_2$ (right). Upper panels: maximum over the angles and parameter sets of Table II for the series through β_A^4 (circles) and through β_A^6 (squares), with lines proportional to β_A^6 and β_A^8. Lower panels: the same maxima divided by β_A^6 and β_A^8.

the maximum comes from the same point at every velocity and for both coefficients, $c = 0$ with $m_A = 150$ GeV, $\mu_R = 75$ GeV and $N_c = 4$. These curves therefore follow the error of a single point as β_A varies, and the slopes and scaled errors discussed below are not affected by a change of the point that gives the maximum. At each velocity and for both orders the errors at the other points are smaller by at most a factor of eight, so the maximum is not set by an isolated point.

For a series in β_A^2, the first omitted terms are of order β_A^6 and β_A^8, up to powers of $\ln\beta_A$; the reference lines in the upper panels show these powers. Straight-line fits of the logarithm of the maximum error against $\ln\beta_A$ over the plotted range give slopes of 6.005 and 8.005 for $\mathcal{M}_1$, and 5.955 and 8.034 for $\mathcal{M}_2$, for the series through β_A^4 and β_A^6.

The lower panels divide the maximum errors by the expected powers, β_A^6 for the series through β_A^4 and β_A^8 for the series through β_A^6, so that exact power behaviour would give horizontal lines. The ratios vary slowly, because both the omitted terms and the remainders that normalise the errors contain powers of $\ln\beta_A$. An incorrect coefficient at a retained order would instead make them grow as an inverse power of β_A, which is not seen anywhere in the tested range. Towards $\beta_A = 0.3$, the scaled errors of the series through β_A^6 rise for both coefficients.

Table III lists the maximum errors at selected velocities.

TABLE II. Validation grid. At each point the six flavour coefficients and the two complete $n_f = 5$ coefficients are compared.

quantity	values
β_A	1/1000, 1/500, 1/200, 1/100, 1/50, 1/20, 1/10, 1/5, 3/10
c	$-2/3, -1/3, 0, 1/3, 2/3$
$(m_A/\text{GeV}, \mu_R/\text{GeV}, N_c)$	(125, 125, 3), (125, 250, 3), (150, 75, 4)

For $\beta_A \le 0.2$ the series through β_A^6 reproduces $\mathcal{M}_1$ and $\mathcal{M}_2$ with relative errors of at most 3.23×10^{-6} and 6.77×10^{-5}; at $\beta_A = 0.3$ the errors are 1.10×10^{-4} and 1.94×10^{-3}. Bose symmetry was checked by comparing the values at c and $-c$: they are identical in all stored digits in 432 comparisons of the unexpanded coefficients and 864 of the series.

The reference values were computed in Mathematica from the published expressions. To check their evaluation independently, I wrote a second code in Python with mpmath [39]. It reads the published files with its own parser and computes $\text{Li}_{2,2}$ from

TABLE III. Maximum relative errors of the complete Type II coefficients on the validation grid. At each value of β_A, the maximum is taken over the 5 angles and 3 parameter sets of Table II.

	$\mathcal{M}_1$ through		$\mathcal{M}_2$ through	
β_A	β_A^4	β_A^6	β_A^4	β_A^6
0.01	1.09×10^{-12}	1.09×10^{-16}	1.87×10^{-11}	2.16×10^{-15}
0.05	1.52×10^{-8}	3.81×10^{-11}	2.63×10^{-7}	8.55×10^{-10}
0.10	1.01×10^{-6}	1.01×10^{-8}	1.63×10^{-5}	2.33×10^{-7}
0.20	7.97×10^{-5}	3.23×10^{-6}	1.02×10^{-3}	6.77×10^{-5}
0.30	1.21×10^{-3}	1.10×10^{-4}	1.15×10^{-2}	1.94×10^{-3}

TABLE IV. Relative truncation errors $E_i^{(6)}$ of the complete $n_f = 5$ coefficients towards the beam, at $m_A = \mu_R = 125\,\text{GeV}$ and $N_c = 3$. The last rows give the largest value on the grid of Table II. The values at $c = -0.99$ equal those at $-c$.

	c	$\beta_A = 0.1$	$\beta_A = 0.2$	$\beta_A = 0.3$
$\mathcal{M}_1$	0.9	6.36×10^{-9}	1.11×10^{-6}	1.78×10^{-5}
$\mathcal{M}_1$	0.99	1.06×10^{-8}	2.43×10^{-6}	5.76×10^{-5}
$\mathcal{M}_1$	0.999	1.23×10^{-8}	2.96×10^{-6}	7.18×10^{-5}
$\mathcal{M}_2$	0.9	1.39×10^{-7}	3.08×10^{-5}	6.68×10^{-4}
$\mathcal{M}_2$	0.99	6.68×10^{-8}	1.47×10^{-5}	2.88×10^{-4}
$\mathcal{M}_2$	0.999	6.21×10^{-8}	1.65×10^{-5}	3.74×10^{-4}
$\mathcal{M}_1$	grid max.	1.01×10^{-8}	3.23×10^{-6}	1.10×10^{-4}
$\mathcal{M}_2$	grid max.	2.33×10^{-7}	6.77×10^{-5}	1.94×10^{-3}

its defining double sum, or from a one-dimensional integral representation where the sum converges slowly. At four points covering the three parameter sets, the two complete coefficients agree with the reference values to a relative difference of at most 1.3×10^{-40}, the precision to which the reference values are stored. The six flavour coefficients, compared at $\beta_A = 1/10$ and $c = 1/3$, agree to 1.6×10^{-40}. At $c = 0$, where $t = u$, the published expressions have removable singularities and are evaluated as limits.

The expansion is made at fixed angle, but p_T also vanishes as $c \to \pm1$ at fixed β_A, so the series has to be tested close to the beam as well. I therefore compared it with the unexpanded remainders at $c = 0.9, 0.99, 0.999$ and -0.99 for $\beta_A = 0.1, 0.2$ and 0.3. Table IV gives the relative errors of the series through β_A^6. All values are below the maximum on the main grid at the same β_A, except for $\mathcal{M}_1$ at $\beta_A = 0.1$, where the largest value, 1.23×10^{-8} at $c = 0.999$, exceeds it by 21%. The series thus remains accurate close to the beam, consistent with its coefficients having no poles at $c = \pm1$, only logarithms of $1 \pm c$ (Sec. IV A).

IV. LEADING POWER AND THE DOUBLE PARTON SCATTERING SINGULARITY

A. Leading power

To determine the leading threshold behaviour of the two-loop remainders, I first collect the terms of order β_A^0, including all powers of $\ln\beta_A$ that multiply them. The notation $\big|_{\beta_A^0}$ used below denotes this part of the expansion, rather than the value of the remainder at $\beta_A = 0$.

In the explicit result, the logarithms of β_A and the angular logarithms combine through $\ln\beta_A + \frac{1}{2}\ln(1-c^2)$. It is convenient to introduce

$$\begin{aligned}\mathcal{L} &= \ln\beta_A + \tfrac{1}{2}\ln(1-c^2) + \ln 2 + \frac{i\pi}{2}\\ &= \tfrac{1}{2}\ln\bigl[-4\beta_A^2(1-c^2)\bigr],\end{aligned} \tag{37}$$

where the second equality uses the branch $\ln(-X) = \ln X + i\pi$ for $X > 0$.

The relation of $\mathcal{L}$ to the physical kinematics follows from Eq. (13). Substituting the expressions for $tu - m_A^4$ and p_T^2 gives

$$\begin{aligned}\mathcal{L} &= \tfrac{1}{2}\ln\frac{m_A^4 - tu}{m_A^4} + \ln(1-\beta_A^2)\\ &= \ln\frac{2p_T}{m_A} + \frac{i\pi}{2} + \tfrac{1}{2}\ln(1-\beta_A^2).\end{aligned}$$

Since $\ln(1-\beta_A^2) = O(\beta_A^2)$, the last term contributes only beyond leading power. Thus, near threshold, the combination of velocity and angular logarithms in Eq. (37) is the logarithm of the transverse momentum, up to the constant phase $i\pi/2$ and power corrections.

When the calculated leading power terms are rewritten in terms of $\mathcal{L}$, no separate $\ln(1\pm c)$ or $\mathrm{Li}_2\bigl((1+c)/2\bigr)$ remains. The rational dependence on c also cancels. The angular dependence is therefore entirely contained in $\mathcal{L}$, while the coefficients of its powers are independent of the scattering angle. For general N_c and n_f, the result is

$$\begin{aligned}\mathcal{M}_1^{(2)}\big|_{\beta_A^0} &= m_A^2\sum_{k=0}^{3}\widehat{A}_{1k}\,\mathcal{L}^k,\\ \mathcal{M}_2^{(2)}\big|_{\beta_A^0} &= m_A^2\sum_{k=0}^{2}\widehat{A}_{2k}\,\mathcal{L}^k.\end{aligned} \tag{38}$$

The highest logarithmic coefficients are

$$\begin{aligned}\widehat{A}_{13} &= -\tfrac{28}{3}\pi N_c^2, \qquad \widehat{A}_{12} = -\tfrac{14}{3}\pi N_c(11N_c - 2n_f),\\ \widehat{A}_{22} &= -2\pi N_c^2, \qquad \widehat{A}_{21} = -\tfrac{4}{3}\pi N_c(4N_c - n_f),\end{aligned} \tag{39}$$

which do not depend on μ_R, and

$$\widehat{A}_{11} = \frac{\pi N_c}{9}\bigl(N_c\,a_\mathrm{A} - n_f\,a_\mathrm{F}\bigr), \tag{40}$$

where

$$\begin{aligned}a_\mathrm{A} &= 1454 + 462\ln 2 + 252\ln^2 2 - 42\pi^2 + 693\,i\pi - 693L_\mu,\\ a_\mathrm{F} &= 152 + 84\ln 2 + 126\,i\pi - 126L_\mu,\end{aligned}$$

TABLE V. Coefficients of the leading power term, $\mathcal{M}_i|_{\beta_A^0} = m_A^2 \sum_k \widehat{A}_{ik}\, \mathcal{L}^k$, with $\mathcal{L}$ of Eq. (37). They do not depend on the scattering angle. Exact forms for $\mu_R = m_A$ and general N_c, n_f; numerical values for $N_c = 3$, $n_f = 5$, $\mu_R = m_A$.

	k	exact	Re $\widehat{A}_{ik}$	Im $\widehat{A}_{ik}$
$\mathcal{M}_1$	3	$-\frac{28}{3}\pi N_c^2$	-2.63894×10^2	0
$\mathcal{M}_1$	2	$-\frac{14}{3}\pi N_c(11N_c - 2n_f)$	-1.01159×10^3	0
$\mathcal{M}_1$	1	Eq. (40)	3.55129×10^3	4.76702×10^3
$\mathcal{M}_1$	0	–	-3.98215×10^2	-4.00569×10^3
$\mathcal{M}_2$	2	$-2\pi N_c^2$	-5.65487×10^1	0
$\mathcal{M}_2$	1	$-\frac{4}{3}\pi N_c(4N_c - n_f)$	-8.79646×10^1	0
$\mathcal{M}_2$	0	–	3.12007×10^2	3.40501×10^2

TABLE VI. Behaviour of the Type II coefficients at threshold, $\beta_A \to 0$ at fixed angle: tree-level values, and the highest power of $\mathcal{L}$ at one and two loops.

	tree level	one loop	two loops
$\mathcal{M}_1$ ($J_z = 0$)	$2im_A^2$	$\mathcal{L}$	$\mathcal{L}^3$
$\mathcal{M}_2$ ($\lvert J_z\rvert = 2$)	$2im_A^2\beta_A^2(1-c^2)$	$\mathcal{L}^0$	$\mathcal{L}^2$

and $L_\mu = \ln(m_A^2/\mu_R^2)$. Table V gives all coefficients for $N_c = 3$, $n_f = 5$ and $\mu_R = m_A$. At β_A^2, β_A^4 and β_A^6 the same substitution again removes every $\ln(1 \pm c)$ and Li_2; the remaining dependence on the angle is rational, even in c and free of poles at $c = \pm 1$.

Up to the constant $i\pi/2$ and power corrections, $\mathcal{L}$ is the logarithm of the transverse momentum of the pseudoscalars or, equivalently, of the Gram determinant $tu - m_A^4$. The exact one-loop remainders already contain it, since they depend on $\ln(y-x)+\ln(z-x)$ and $(y-x)(z-x) = yz-1 = (tu-m_A^4)/m_A^4$ by Eq. (10). The two-loop coefficients have no finite limit as $\beta_A \to 0$; they grow as $\mathcal{L}^3$ and $\mathcal{L}^2$.

B. One loop

Fitting the published one-loop remainders at $\beta_A = 10^{-40}$ and 10^{-50} with 400-digit arithmetic, for $N_c = 3$, 4 and 5, $n_f = 0$, 2 and 5, two values of μ_R and several angles, I find to 30 digits

$$\begin{aligned} \mathcal{M}_1^{(1)} &= m_A^2\big[7\pi N_c\,\mathcal{L} + \widehat{A}_{10}^{(1)}\big] + O(\beta_A), \\ \mathcal{M}_2^{(1)} &= \tfrac{1}{2}\pi N_c\, m_A^2 + O(\beta_A), \end{aligned} \tag{41}$$

with $\widehat{A}_{10}^{(1)} = -93.900 + 11.484\,i$ for $N_c = 3$, $n_f = 5$ and $\mu_R = m_A$, in the normalisation of Eq. (23). The constant does not depend on the angle, and by Eq. (23) its scale dependence is $\partial \widehat{A}_{10}^{(1)}/\partial \ln \mu_R^2 = 4i\beta_0$. The coefficient of $\mathcal{L}$ and the limit of $\mathcal{M}_2^{(1)}$ depend neither on n_f nor on μ_R. Together with Eqs. (19) and (38), the behaviour at threshold is summarised in Table VI.

The $|J_z| = 2$ amplitude is suppressed near threshold only at tree level. According to Eq. (18) the helicity amplitude $\mathcal{A}_{\pm\mp}$ then keeps its dependence $e^{\pm 2i\phi}$ on the azimuth of $\vec{p}_3$ as $|\vec{p}_3| \to 0$. An amplitude that is analytic in $\vec{p}_3$ at $\vec{p}_3 = 0$ cannot behave in this way; the loop amplitudes can, because they depend on $\ln p_T$.

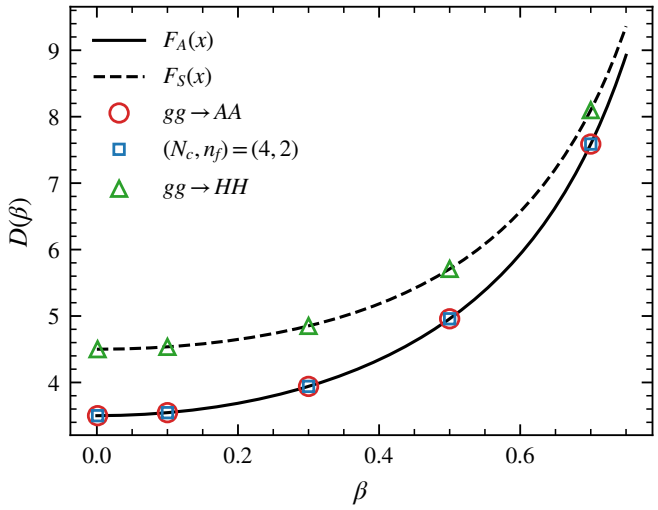


FIG. 3. Forward coefficient $D(\beta)$ of $\ln p_T^2$ in the one-loop remainder, Eq. (44), against the velocity β of the produced bosons. Circles: $gg \to AA$ with $(N_c, n_f) = (3, 5)$; squares: $(N_c, n_f) = (4, 2)$; the two coincide. Triangles: $gg \to HH$ through two CP-even vertices at $(N_c, n_f) = (3, 5)$, multiplied by -1, the sign of its tree amplitude at threshold relative to AA. Lines: the helicity weights F_A (solid) and F_S (dashed) of Eq. (43) at $x = (1+\beta)/2$.

C. The double parton scattering singularity

The origin of $\mathcal{L}$ can be identified. At one loop the Type II amplitude contains diagrams in which each incoming gluon splits into two gluons through a three-gluon vertex, and the four gluons fuse pairwise at the two effective vertices into the pseudoscalars. When $p_T \to 0$ the four internal propagators can go on shell simultaneously, with one pair collinear to each incoming gluon. This configuration gives the pinch singularity that Gaunt and Stirling identified as the DPS singularity of one-loop amplitudes [2]. For theories of the Standard Model type they showed that it is at most logarithmic in p_T, that in the processes they considered, with fermions in the loop, it appears only if the incoming partons are in a state with $J_z = 0$, and that

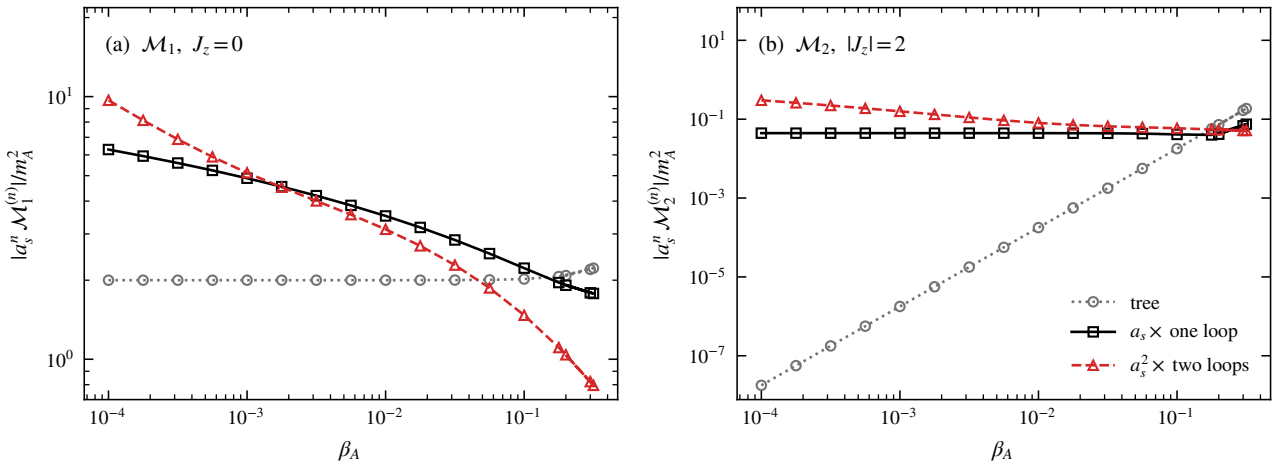


FIG. 4. Tree-level, one-loop and two-loop terms of the Type II amplitude, $|a_s^n \mathcal{M}_i^{(n)}|/m_A^2$ with $a_s = 0.0094$ ($\alpha_s = 0.118$), at $c = 1/3$, $m_A = \mu_R = 125\,\text{GeV}$, $N_c = 3$ and $n_f = 5$. Tree level: Eq. (19); one loop: published remainder; two loops: series through β_A^6. $\mathcal{M}_1$ and $\mathcal{M}_2$ are the equal and opposite helicity amplitudes, Eq. (18).

its coefficient is a transverse momentum integral of light-cone wavefunctions of the incoming partons and tree-level matrix elements of the two $2 \to 1$ subprocesses. In their analysis the strength of the singularity is set by the vertices at which the incoming partons split, whereas the $2 \to 1$ amplitudes enter only through their values for collinear on-shell partons at zero transverse momentum. Here the splittings occur at ordinary three-gluon vertices and the effective vertices of O_G enter as the amplitudes for $gg \to A$, so the analysis applies. For $g \to gg$ every collinear splitting changes J_z, since the total helicity of a collinear pair of gluons is ± 2 or 0, never that of the parent gluon, so the splitting amplitudes vanish linearly in the transverse momentum and the singularity is logarithmic, as for loops of massless quarks [2]. With gluons in the loop, helicity conservation no longer restricts the four partons of the singular configuration to $J_z = 0$, and the requirement is conservation of J_z between the incoming partons and these four partons [3]. For the present process this again selects $J_z = 0$. The amplitude for $gg \to A$ vanishes for opposite gluon helicities, so each pair of gluons that fuses at an effective vertex has $J_z = 0$, and so does the configuration as a whole; the singularity can therefore appear only for incoming gluons with $J_z = 0$.

In the present case the subprocesses are $gg \to A$ through O_G. Their amplitudes vanish for opposite gluon helicities, and, since a parity transformation reverses all helicities [40], a CP-odd coupling gives amplitudes of opposite sign for $g^+g^+ \to A$ and $g^-g^- \to A$; for a scalar coupling to G^2 they would be equal. For the splitting of a gluon of positive helicity into gluons a and b with momentum fractions z and $1-z$, the splitting amplitudes are proportional to [40, 41]

$$\frac{1}{\sqrt{z(1-z)}}, \quad \frac{z^2}{\sqrt{z(1-z)}}, \quad \frac{(1-z)^2}{\sqrt{z(1-z)}} \tag{42}$$

for the helicities $(h_a, h_b) = (+,+)$, $(+,-)$ and $(-,+)$; the sum of their squares gives the numerator $1 + z^4 + (1-z)^4$ of the $g \to gg$ splitting function. Let the first pseudoscalar receive the fraction x of the momentum of the first gluon and the fraction $1-x$ of the momentum of the second, as for the kinematics of Ref. [2] with equal masses; in Fig. 1(b) this is the point $k = xp_1$, $p_3 - k = (1-x)p_2$. Summing over the three helicity configurations allowed by the subprocesses, with the signs of their amplitudes, gives the weight

$$F_A(x) = \frac{1}{x(1-x)} - 2x(1-x), \tag{43}$$

where the first term comes from $(+,+)$ splittings of both gluons and the second from the two mixed configurations, in which one effective vertex receives two gluons of positive and the other two of negative helicity, so that this term carries the relative sign of the two amplitudes. A scalar would have $F_S(x) = 1/[x(1-x)] + 2x(1-x)$ instead. Boxes of massless quarks have a different weight: for $gg \to HH$ through a quark loop the prefactor is proportional to $x(1-x)$, because the internal lines of equal helicity carry complementary momentum fractions [2].

At threshold both pseudoscalars are at rest and $x = \frac{1}{2}$, so that $F_A = \frac{7}{2}$ and $F_S = \frac{9}{2}$. The coefficient $7\pi N_c$ of $\mathcal{L}$ in Eq. (41) corresponds to $\partial \mathcal{M}_1^{(1)}/\partial \ln p_T^2 = \frac{7}{2}\pi N_c m_A^2$, that is $\pi N_c m_A^2 F_A(\frac{1}{2})$. A test that does not rely on the overall factor is provided by the forward limit $c \to 1$ at fixed velocity, where $p_T \to 0$ as well but the bosons move along the beam. Writing β and m for the velocity and mass of the produced bosons ($\beta = \beta_A$ and $m = m_A$ for AA), so that $x = (1+\beta)/2$, the DPS singularity requires for the forward coefficient $D(\beta)$

$$D(\beta) \equiv \frac{1}{\pi N_c m^2} \frac{\partial\ \text{Re}\,\mathcal{M}_1^{(1)}}{\partial \ln p_T^2}\bigg|_{c\to 1} = F_A\left(\frac{1+\beta}{2}\right). \tag{44}$$

I evaluated D from the exact one-loop remainders at $1 - c = 10^{-14}$ and 10^{-18}, for $0.001 \le \beta \le 0.7$, $N_c = 3$ and 4 and $n_f = 0$, 2 and 5. Equation (44) holds in all cases with a largest relative deviation of 3×10^{-15} (Fig. 3). The weight of a scalar is excluded at every $\beta > 0$. The overall factor $\pi N_c m^2$ is fixed by the threshold value; I have not derived it from the vertices.

The identification explains the other features of Eq. (41). The $|J_z| = 2$ amplitude $\mathcal{M}_2$ has no logarithm at one loop, as the J_z argument above requires. The coefficient of $\mathcal{L}$ does not depend on n_f, because the vertices of O_G couple to gluons only, so that only gluon splittings enter the singular configuration. It does not depend on μ_R, because the singular term is finite and receives no contribution from renormalisation or from the infrared subtraction. Its colour factor N_c is that of the two three-gluon vertices. Finally, the coefficient $7\pi N_c\, m_A^2$ is real while the tree amplitude $\mathcal{M}_1^{(0)} = 2im_A^2$ is imaginary, so the logarithm is out of phase with the tree amplitude by 90°. This holds for any real couplings: the DPS singularity of a one-loop amplitude comes from the cuts that put the internal lines on shell and is therefore absorptive [2]. This has direct consequences for the cross section (Sec. V).

A CP-even coupling, $HG^a_{\mu\nu}G^{a\mu\nu}$, gives equal amplitudes for $g^+g^+ \to H$ and $g^-g^- \to H$, which reverses the sign of the mixed term and gives the weight F_S. The topology, the splittings and the colour factors are the same as for AA, so HH tests the part of the formula that depends on the vertices. I tested this prediction with the one-loop Type II amplitudes for $gg \to HH$ at the same mass m, computed with the code of Ref. [7] after replacing the effective vertices by the CP-even ones; the code reproduces the one-loop AA result of that reference to 40 digits, and the HH double poles are those of Catani's formula [30]. The HH tree amplitude at threshold, $-2im^2$, has the sign opposite to that of AA, and $D(\beta) = -F_S\big((1+\beta)/2\big)$ to 3×10^{-15} for $0.001 \le \beta \le 0.7$ (Fig. 3). In units of the tree amplitude, the normalisation fixed from AA thus predicts the HH logarithm without a free parameter, in the ratio $F_S(\frac{1}{2})/F_A(\frac{1}{2}) = \frac{9}{7}$ at threshold. The $|J_z| = 2$ amplitude again has no logarithm; its forward coefficient is below 3×10^{-14} of that of $\mathcal{M}_1$. The diagram with the four-gluon term of the CP-even operator and the contact counterterm of two such vertices, neither of which depends on p_T, were not included.

D. Two loops

At two loops the leading logarithms of $\mathcal{M}_1$ are related to the one-loop DPS logarithm,

$$\widehat{A}_{13}\mathcal{L}^3 + \widehat{A}_{12}\mathcal{L}^2 = 7\pi N_c\,\mathcal{L}\left[-\tfrac{4N_c}{3}\,\mathcal{L}^2 - 2\beta_0\,\mathcal{L}\right], \qquad (45)$$

with $\beta_0 = (11N_c - 2n_f)/3$. In planar $\mathcal{N} = 4$ super Yang–Mills theory the singular terms of the MHV six-gluon amplitude near the configuration that mimics double parton scattering obey an evolution equation, and the leading logarithms are known to all orders [3]. There the ratio of the two- and one-loop leading logarithms is $-(a/6)\ln^2\delta$, with $a = g^2N_c/(8\pi^2)$ and δ proportional to the square of the transverse momentum. With $a = 2N_c a_s$ and $\ln\delta = 2\mathcal{L} + \text{const}$ this is $-\frac{4}{3}N_c a_s \mathcal{L}^2$, the ratio of the first terms in Eq. (45). The second term, $-2\beta_0\mathcal{L}$, has no counterpart there, since the β function of that theory vanishes, and the number of quark flavours enters it only through β_0. The same β_0 term appears in the momentum-space kernel for the splitting $g \to gg$ in unpolarised colour singlet double parton distributions at next-to-leading order [22], where the transverse momentum mismatch Δ between amplitude and conjugate plays the role of p_T. Relative to the leading order term, the β_0 part of the coefficient of $\ln^2(\mu^2/\Delta^2)$ corresponds to $-2\beta_0\mathcal{L}$. One half comes from the renormalisation of the coupling and the other half from the δ-function parts of the DGLAP kernels, so that the running of the coupling alone would give $-\beta_0\mathcal{L}$. The remaining terms, one proportional to $C_A\ln[x(1-x)]$ from the plus distributions, which contributes where the two momentum fractions add up to one, and the real emission terms, which contribute where they add up to less than one, have no counterpart in the amplitude. The $|J_z| = 2$ amplitude acquires a logarithm at two loops, $\widehat{A}_{22} = (-4N_c)\cdot\frac{1}{2}\pi N_c$, that is the one-loop constant multiplied by $-4N_c\,\mathcal{L}^2$. The J_z argument of Sec. IV C concerns the one-loop singularity and does not forbid this. The logarithm cannot come from two-loop amplitudes of the subprocesses, since conservation of angular momentum forbids $gg \to A$ for opposite gluon helicities to all orders, so the one-loop mechanism does not explain it; I leave its origin open. A related effect occurs in planar $\mathcal{N} = 4$ super Yang–Mills theory: the NMHV six-gluon amplitude, which has no such singularity at one loop because of J_z conservation, has logarithmic singularities from two loops on, found where the limit overlaps with multi-Regge kinematics, and Dixon and Esterlis suggest that they may come from contributions in which more than one particle crosses a cut [3].

Figure 4 compares the three orders. With $a_s = 0.0094$ ($\alpha_s = 0.118$), at $\beta_A = 0.1$ and $c = 1/3$, the one- and two-loop terms of $\mathcal{M}_1$ are 1.1 and 0.7 times the tree term, and the one-loop term of $\mathcal{M}_2$ is 2.3 times the tree term. The last ratio grows as β_A^{-2} towards threshold, since the tree-level $|J_z| = 2$ amplitude vanishes there. The one-loop terms include the scale term of Eq. (23). Towards threshold the loop terms of $\mathcal{M}_1$ grow with $|\mathcal{L}|$, the two-loop term as $|\mathcal{L}|^3$ and the one-loop term as $|\mathcal{L}|$, so that below $\beta_A \approx 1.7\times10^{-3}$ the two-loop term exceeds the one-loop term at this angle. Because the logarithms are out of phase with the tree amplitude, this growth largely drops out of the cross section (Sec. V). In panel (b) the one-loop term of $\mathcal{M}_2$ tends to the constant of Eq. (41), while the two-loop term grows as $\mathcal{L}^2$, the logarithm that the $|J_z| = 2$ amplitude acquires at two loops.

V. CONSEQUENCES FOR THE CROSS SECTION

A. Leading order

Averaging Eq. (18) over helicities and colours, and including a factor $\frac{1}{2}$ for identical particles, gives the leading order partonic cross section of the Type II amplitude,

$$\frac{\mathrm{d}\hat\sigma}{\mathrm{d}c} = \frac{\beta_A\,C_G^4}{128\pi\hat s\,(N_c^2-1)}\left(|\mathcal{M}_1^{(0)}|^2 + |\mathcal{M}_2^{(0)}|^2\right) \qquad (46)$$

for $-1 < c < 1$. For $\alpha_s = 0.118$, $\cot\beta_H = 1$ and $m_A = 125$ GeV I find $\hat\sigma = 1.29\times10^{-8}$ pb at $\beta_A = 0.1$ (Fig. 5(a)). The cross section rises as β_A above threshold and as $\hat s$ at large $\hat s$, as expected for dimension-five operators, and it is of fourth order

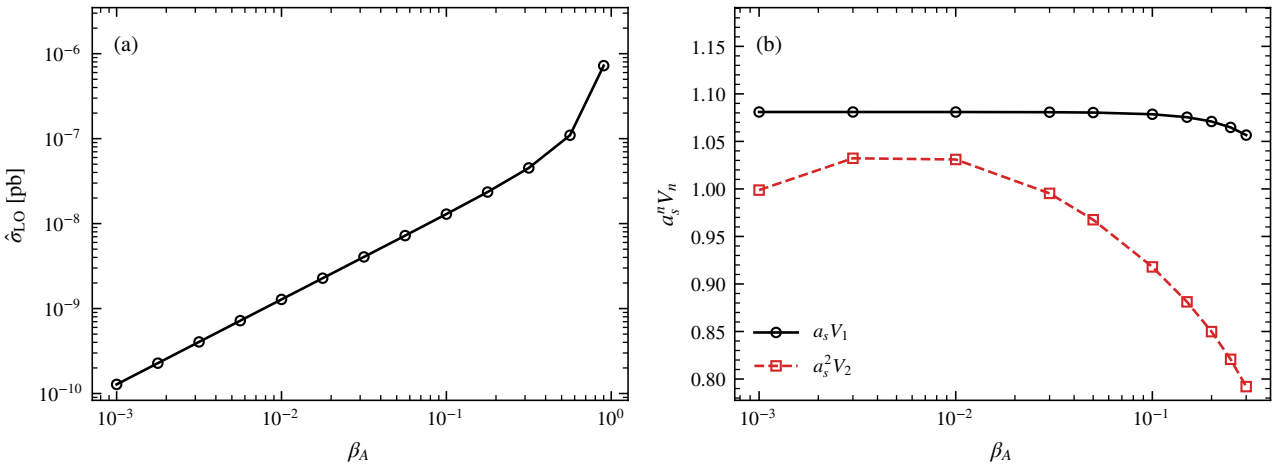


FIG. 5. (a) Leading order partonic cross section of the Type II amplitude, Eq. (46), for $m_A = 125\,\text{GeV}$, $\alpha_s = 0.118$ and $\cot\beta_H = 1$. (b) Finite virtual corrections of Eq. (47), averaged over the angle, for $\mu_R = m_A$, $N_c = 3$ and $n_f = 5$.

in a_s through C_G^4. In a two-Higgs-doublet model it is a small part of the total, which is dominated by Type I (Sec. II E).

B. Finite virtual corrections

With all remainders in one normalisation,

$$\sum |\mathcal{A}|^2 \propto \sum_i |\mathcal{M}_i^{(0)}|^2 \left[1 + a_s V_1 + a_s^2 V_2\right],$$
$$V_1 = \frac{2\,\text{Re}\sum_i \mathcal{M}_i^{(0)*}\mathcal{M}_i^{(1)}}{\sum_i |\mathcal{M}_i^{(0)}|^2},$$
$$V_2 = \frac{\sum_i \left(2\,\text{Re}\ \mathcal{M}_i^{(0)*}\mathcal{M}_i^{(2)} + |\mathcal{M}_i^{(1)}|^2\right)}{\sum_i |\mathcal{M}_i^{(0)}|^2}. \qquad (47)$$

I average over the angle with the leading order weight and evaluate $\mathcal{M}^{(1)}$ exactly and $\mathcal{M}_i^{(2)}$ from the series through β_A^6, both in the normalisation of Eq. (23). For $\alpha_s = 0.118$ and $\mu_R = m_A = 125\,\text{GeV}$, $a_s V_1$ lies between 1.06 and 1.08 and $a_s^2 V_2$ between 0.79 and 1.03 for $10^{-3} \le \beta_A \le 0.3$ (Fig. 5(b)). V_1 and V_2 are finite parts after the subtraction of Eq. (21) and depend on that scheme; they are neither observables nor K factors. By Eq. (23), V_1 depends on the scale as $\partial V_1/\partial \ln\mu_R^2 = 4\beta_0$.

Near threshold the logarithms of Sec. IV enter the cross section in a simple way. Since $\mathcal{M}_1^{(0)} \to 2im_A^2$ and $|\mathcal{M}_2^{(0)}| \propto \beta_A^2$, Eq. (41) gives

$$V_1 \;\longrightarrow\; \frac{\pi}{2}\cdot 7\pi N_c + \text{Im}\,\widehat{A}_{10}^{(1)} = 115.11 \qquad (48)$$

for $\mu_R = m_A$. The DPS logarithm drops out, because it is out of phase with the tree amplitude, and only the constant $i\pi/2$ contained in $\mathcal{L}$ contributes, together with the constant $\widehat{A}_{10}^{(1)}$. V_1 therefore approaches a constant, as seen in Fig. 5. At two loops, write $\mathcal{L} = \ell + i\pi/2$ with $\ell = \ln(2p_T/m_A)$. At fixed angle the leading power terms give

$$V_2 = v_0 + v_1\,\ell + v_2\,\ell^2, \qquad (49)$$

and ℓ^3 cancels because $\widehat{A}_{13}$ is real; $\widehat{A}_{12}$ is real as well and does not contribute to ℓ^2. The coefficient of ℓ^2 receives $\frac{3}{2}\pi\widehat{A}_{13} = -14\pi^2 N_c^2$ from the interference of the tree and two-loop amplitudes and $\frac{1}{4}(7\pi N_c)^2 = \frac{49}{4}\pi^2 N_c^2$ from the square of the one-loop amplitude, so that

$$v_2 = -\tfrac{7}{4}\pi^2 N_c^2. \qquad (50)$$

A fit of V_2 at $c = 1/3$ and $10^{-8} \le \beta_A \le 10^{-6}$ gives $v_2 = -155.446269$, against -155.446269 from Eq. (50), a relative difference of 3×10^{-11}. The DPS logarithms thus reach the finite virtual corrections only at two loops, where $\ln^2 p_T$ is the highest power; with $a_s = 0.0094$, $a_s^2 v_2 = -0.0137$. In the language of double parton scattering the two contributions to v_2 are of different kinds. The square $|\mathcal{M}_i^{(1)}|^2$ has the DPS region on both sides of the final-state cut, like the contribution with a splitting in both protons that overlaps with double parton scattering and is removed from it by a subtraction [23, 27]. The interference with the tree amplitude has the DPS region on one side only. It carries no logarithm at one loop, because the DPS singular part is absorptive, and at two loops it is the larger of the two and fixes the sign of v_2. At leading power v_0 and v_1 do not depend on the angle either, since the angle enters only through ℓ; fits at $c = 0$, $\frac{1}{3}$ and $\frac{2}{3}$ agree to 5×10^{-11}. Equations (39)–(41) give $v_1 = \pi\widehat{A}_{12} + \text{Im}\,\widehat{A}_{11} + \frac{7}{2}\pi N_c\,\text{Re}\,\widehat{A}_{10}^{(1)} = \frac{7}{2}\pi N_c\,\text{Re}\,\widehat{A}_{10}^{(1)} + 7\pi^2 N_c\beta_0 = -1508.5$, in agreement with the fit to 1×10^{-10}. It does not depend on μ_R, because $\text{Re}\,\widehat{A}_{10}^{(1)}$ does not; the scale dependence of V_2 sits in v_0, which changes from 8118 to 15421 when μ_R is doubled. Although ℓ^2 is the highest power, it is not the largest term at these velocities: at $c = 1/3$ the logarithmic terms $v_1\ell + v_2\ell^2$ make up 20% of V_2 at $\beta_A = 0.1$ and 30% at $\beta_A = 0.01$, of which $v_2\ell^2$ contributes -4% and -21%. Since v_1 and v_2 are both negative, V_2 has a maximum at $\ell = -v_1/(2v_2) = -4.85$, that is at $\beta_A = 0.0041$ for $c = 1/3$, and decreases above it; the angle averaged $a_s^2 V_2$ of Fig. 5(b) falls for the same reason. The terms beyond leading power change V_2 by only 0.3% at $\beta_A = 0.1$. The coefficient v_2 does not depend on the conventions of the infrared subtraction: a change

of the finite parts of the operators in Eq. (21), which depend on $\hat{s}$ only, shifts the remainders by lower order amplitudes multiplied by angle independent coefficients, and so changes only v_0 and v_1.

The one-loop O_G–O_J diagrams contribute to the cross section at the same order as V_2, through the physical coefficient C_J, which depends on m_t [7]. By Eq. (24) this contribution is proportional to the tree amplitude, so it shifts V_2 by a constant; it is not included here.

C. Use in higher order calculations

Since $\sqrt{\hat{s}}/(2m_A) = (1-\beta_A^2)^{-1/2}$, the range $\beta_A \leq 0.2$ covers the first 2.06% above threshold in $\sqrt{\hat{s}}$, that is $\sqrt{\hat{s}} \leq 255.2\,\text{GeV}$ for $m_A = 125\,\text{GeV}$; $\beta_A = 0.3$ lies 4.83% above threshold. I recommend the series through β_A^6 for $\beta_A \leq 0.2$ at the angles tested, those of Table II and the forward angles of Table IV; the logarithms diverge at $c = \pm 1$, where no bound has been established. At $\beta_A = 0.3$ the series still reproduce both coefficients to 1.94×10^{-3}. The series retain the complex values of both tensor coefficients and can be combined coherently with other contributions once normalisation, Wilson coefficients and tensor basis are matched.

A threshold resummed or soft-virtual prediction of the invariant mass distribution of the pair requires, in addition to universal soft and collinear functions, the hard function of the process at each value of m_{AA}: the infrared subtracted virtual amplitude, weighted with the tree amplitude and averaged over the angle [16–18]. For Higgs pairs this is the construction of Ref. [19], in which the two-loop virtual correction enters integrated over t at fixed invariant mass. The soft and collinear functions of gluon fusion are the same for any colour singlet final state [20]. For m_{AA} within a few per cent of $2m_A$ the present series give the Type II two-loop contribution to this hard function in closed form. It enters at the order at which the remainders enter the cross section (Sec. II E): at NNLO for a pseudoscalar that couples to gluons only through $G\tilde{G}$, and at N^{3}LO in a two-Higgs-doublet model. For Higgs pairs, likewise, the NNLO virtual corrections contain the diagrams with two effective vertices only at tree level and at one loop [42], and their two-loop terms are an ingredient of N^{3}LO [9].

VI. CONCLUSIONS

I have expanded the two-loop Type II finite remainders for $gg \to AA$ about the pair threshold through β_A^6 at fixed scattering angle. After the endpoint constants are expressed in a basis of eight transcendental constants, all inverse powers of β_A and the odd powers through β_A^5 cancel identically, and every coefficient is even in $\cos\theta$. The series reproduce the unexpanded remainders to 3.23×10^{-6} and 6.77×10^{-5} for $\beta_A \leq 0.2$, including the forward angles tested. They are faster to evaluate than the published expressions and 5061 times smaller than the published files. I also found that the released two-loop remainders are larger by a factor of 64 than the tree and one-loop amplitudes, and that the released one-loop remainders omit their dependence on the renormalisation scale. With both corrected, the one- and two-loop remainders satisfy the renormalisation group relations exactly.

At leading power the angular dependence enters only through the logarithm of the transverse momentum of the pseudoscalars. I identified this logarithm with the double parton scattering singularity of Gaunt and Stirling. At one loop the velocity dependence of its coefficient follows their factorised formula, with the helicity weights of gluon splitting and the sign structure of a CP-odd coupling, to 3×10^{-15} over the range of velocities tested, once the overall constant is fixed at threshold. The same identification explains why the $|J_z| = 2$ amplitude has no logarithm at one loop, why the coefficient depends neither on n_f nor on μ_R, and why the logarithm is out of phase with the tree amplitude. At two loops the $J_z = 0$ amplitude grows as $\ln^3 p_T$ and the $|J_z| = 2$ amplitude as $\ln^2 p_T$, and the leading logarithms are simple multiples of the one-loop terms. The ratio of the leading logarithms is the one found to all orders in planar $\mathcal{N} = 4$ super Yang–Mills theory, and the next logarithm has the β_0 structure of the next-to-leading order splitting kernel for $g \to gg$. In the finite virtual corrections to the cross section the logarithm cancels at one loop; at two loops $\ln^3 p_T$ cancels and the highest power is $\ln^2 p_T$, with coefficient $-\frac{7}{4}\pi^2 N_c^2$.

Several questions remain open. The overall normalisation $\pi N_c m^2$ of the one-loop DPS logarithm, which is the same for AA and HH in units of the tree amplitude, should follow from the factorised formula of Ref. [2] with the effective vertices. That formula reproduces the numerical prefactors of the quark boxes, so this is a well-defined calculation. The two-loop logarithms, including those of the $|J_z| = 2$ amplitude, require its extension beyond one loop; the evolution equation of planar $\mathcal{N} = 4$ super Yang–Mills theory [3] and the next-to-leading order splitting kernels of Ref. [22] suggest the form such an extension might take. If the factorisation carries over to the full theory, the top quark mass changes the normalisation of the DPS logarithm by the on-shell vertex factor of Sec. II E, about 1.096 at $m_A = 125$ GeV; its effect on the remaining terms is not known. On the phenomenological side, the contribution of the physical coefficient C_J, which depends on m_t, shifts V_2 by a constant that is not included here, and the series provide the Type II two-loop contribution to the hard function for a threshold resummed description of pseudoscalar pairs near $m_{AA} \approx 2m_A$. For a pseudoscalar that couples to gluons only through $G\tilde{G}$, such as an axion-like particle, they give the O_G–O_G part of the two-loop virtual corrections at NNLO; combined with the soft and collinear functions of Refs. [18, 20], and together with that term, they would give a soft-virtual NNLO prediction for the invariant mass distribution of such particles near threshold. For Higgs pairs produced through two effective vertices, an explicit one-loop calculation confirms the scalar weight F_S, with the constant fixed from AA, and the absence of a one-loop logarithm in the $|J_z| = 2$ amplitude (Sec. IV C). The two-loop amplitudes of this class are known [9], so the two-loop logarithms can be examined as well.

TABLE VII. Reductions of the endpoint constants of Eq. (A1), found with PSLQ at 110 digits; the largest residual is 4×10^{-107}.

$[p,q,a,w]$	reduction
$[1,2,-1,1]$	$-\frac{\pi^2\ln 2}{6}+\frac{5\zeta_3}{8}$
$[1,2,1/2,1/2]$	$-\frac{\zeta_3}{4}-\frac{\ln^3 2}{6}+\frac{\pi^2\ln 2}{12}$
$[1,2,1/2,1]$	$-\zeta_3+\frac{\pi^2\ln 2}{4}$
$[2,1,-1,-1]$	$\frac{\zeta_3}{8}$
$[2,1,-1,1]$	$-\frac{13\zeta_3}{8}+\frac{\pi^2\ln 2}{4}$
$[2,1,1,-1]$	$-\frac{\pi^2\ln 2}{4}+\zeta_3$
$[2,1,1,1/2]$	$-\frac{5\zeta_3}{8}+\frac{\pi^2\ln 2}{6}$
$[2,1,1/2,1/2]$	$-\frac{\ln^3 2}{6}+\frac{\zeta_3}{8}$
$[2,1,1/2,1]$	$-\frac{\pi^2\ln 2}{4}+\frac{13\zeta_3}{8}$
$[2,2,1,-1]$	$-\frac{71\pi^4}{1440}-\frac{\pi^2\ln^2 2}{6}+\frac{\ln^4 2}{6}+4\,\mathrm{Li}_4(\frac{1}{2})+\frac{7\ln 2\zeta_3}{2}$
$[2,2,1,1/2]$	$-\mathrm{Li}_4(\frac{1}{2})-\frac{\ln 2\zeta_3}{4}-\frac{\pi^2\ln^2 2}{24}-\frac{\ln^4 2}{24}+\frac{19\pi^4}{1440}$
$[2,2,1/2,1/2]$	$-\frac{\pi^2\ln^2 2}{24}+\frac{\ln^4 2}{24}+\frac{\pi^4}{1440}+\frac{\ln 2\zeta_3}{4}$
$[2,2,1/2,1]$	$-3\,\mathrm{Li}_4(\frac{1}{2})-\frac{\pi^2\ln^2 2}{8}-\frac{\ln^4 2}{8}+\frac{7\pi^4}{288}$

DATA AVAILABILITY

The threshold series, construction files, validation records, the code of the construction and validation, the evaluator, the one-loop $gg\to HH$ coefficients of Sec. IV C and the scripts that reproduce all tables and figures are provided as ancillary files with the arXiv version of this article and as Supplemental Material [28]. The unexpanded finite remainders used as input were published with Ref. [7]. The code used for the one-loop $gg\to HH$ amplitudes of Sec. IV C is not included; it is available from the author on request.

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I used Claude (Anthropic) to assist with the literature search, with writing and debugging the computer-algebra code of the threshold construction and validation and the analysis and checking scripts, with the figures and with the drafting of the text; I have reviewed and verified all results and text.

Appendix A: Endpoint constants

The boundary values of the expansion contain the constants

$$\mathrm{MB}[p,q,a,w]=\frac{a}{(p-1)!}\int_0^1 \mathrm{d}t\,\frac{[-\ln t]^{p-1}}{1-at}\,\phi_q(wt), \quad \text{(A1)}$$

with $\phi_1(z)=-\ln(1-z)$ and $\phi_q(z)=\mathrm{Li}_q(z)$ for $q\geq 2$. Besides the elementary values $\mathrm{MB}[2,2,1,1]=\pi^4/120$ and $\mathrm{MB}[2,1,1,1]=\zeta_3$, thirteen distinct constants occur. They were evaluated by quadrature at high precision, independently of the amplitude data, and reduced to the basis of Eq. (31) with the PSLQ algorithm [38] at 110 digits. Table VII lists the results; the largest residual is 4×10^{-107}, and the same relations were obtained in Mathematica.

Appendix B: Normalisation of the published remainders

Five checks fix the normalisation of the tree, one-loop and two-loop amplitudes, Eq. (23). They use the tree amplitudes and the one- and two-loop remainders released with Ref. [7], and the Mathematica code of the original calculation.

Tree level. The projections of the two tree diagrams from the original calculation give Eq. (19), after removal of a colour sum N_c^2-1 that the file for T_1 carries and the file for T_2 does not. A direct evaluation of the two diagrams with the vertex $iC_G\varepsilon_{\mu\alpha\rho\beta}k_1^\mu k_2^\rho\delta^{ab}$, which follows from Eq. (1), gives $|\mathcal{A}_{\pm\pm}|=C_G^2|\mathcal{M}_1^{(0)}|$ and $|\mathcal{A}_{\pm\mp}|=C_G^2|\mathcal{M}_2^{(0)}|$ to 14 digits.

Tree and one loop. The coefficient of n_f in $\mathcal{M}_i^{(1)}$, divided by $\mathcal{M}_i^{(0)}$, is the same for $i=1,2$ at threshold, and its imaginary part equals $-2\pi/3$ to 15 digits at every tested point. This is the value produced by the subtraction term $2I^{(1)}\mathcal{M}^{(0)}$ of Eq. (21) through the n_f part of β_0 and $\ln(-\hat{s}/\mu_R^2)$; the one-loop n_f diagrams have no $\hat{s}$-channel cut. It fixes the normalisation and phase of the tree amplitude relative to the one-loop remainders.

Scale dependence at one loop. A remainder with the dimension of m_A^2 can depend on m_A and μ_R only through their ratio, apart from the overall factor m_A^2. The published one-loop remainders violate this: when m_A and μ_R are multiplied by a common factor λ at fixed β_A and c, $\mathcal{M}^{(1)}_{i,\mathrm{publ}}/m_A^2$ changes by $-2\beta_0\ln\lambda^2\,\mathcal{M}_i^{(0)}/m_A^2$, to a relative accuracy of 7×10^{-56}. They are independent of μ_R, since their eight $\ln\mu_R^2$ terms cancel. In the original calculation the amplitude is renormalised with Z_{GG}^2, $Z_{GG}=Z_{a_s}$, and $C_G\propto a_s$; with the subtraction operators of Eq. (21), whose scale dependence enters through $(\mu_R^2/(-\hat{s}))^\epsilon$, renormalisation group invariance of Eq. (20) then requires

$$\frac{\partial\mathcal{M}^{(1)}}{\partial\ln\mu_R^2}=2\beta_0\,\mathcal{M}^{(0)}. \quad \text{(B1)}$$

The first line of Eq. (23) restores this dependence; the corrected remainders are invariant under the rescaling to 6×10^{-57}.

Scale dependence at two loops. With the same ingredients, and with the subtraction operator $I^{(2)}$ of the original code, which, unlike Eq. (3.4) of Ref. [7], does not multiply the constant

$$H_g^{(2)}=\tfrac{1}{2}\Big[C_A^2\big(-\tfrac{5}{12}-\tfrac{11}{24}\zeta_2-\tfrac{1}{2}\zeta_3\big)+C_An_f\big(\tfrac{29}{27}+\tfrac{\zeta_2}{12}\big)-\tfrac{1}{2}C_Fn_f-\tfrac{5}{27}n_f^2\Big] \quad \text{(B2)}$$

by $(-\hat{s}/\mu_R^2)^\epsilon$, one finds

$$\frac{\partial\mathcal{M}^{(2)}}{\partial\ln\mu_R^2}=3\beta_0\,\mathcal{M}^{(1)}+\big(2\beta_1-8H_g^{(2)}-12C_Fn_f\big)\mathcal{M}^{(0)}, \quad \text{(B3)}$$

where $\beta_1=\frac{34}{3}C_A^2-2C_Fn_f-\frac{10}{3}C_An_f$ and the last term comes from the counterterm of the O_G–O_J amplitude, Eq. (24), in the renormalisation of Ref. [7]. I fitted $\partial\mathcal{M}^{(2)}/\partial\ln\mu_R^2=a\,\beta_0\mathcal{M}^{(1)}+b\,\mathcal{M}^{(0)}$ to the series at two velocities and angles, for $(N_c,n_f,m_A)=(3,0,300\,\mathrm{GeV})$, $(3,5,125\,\mathrm{GeV})$ and $(4,2,150\,\mathrm{GeV})$. With both lines of Eq. (23), $a=3$ to

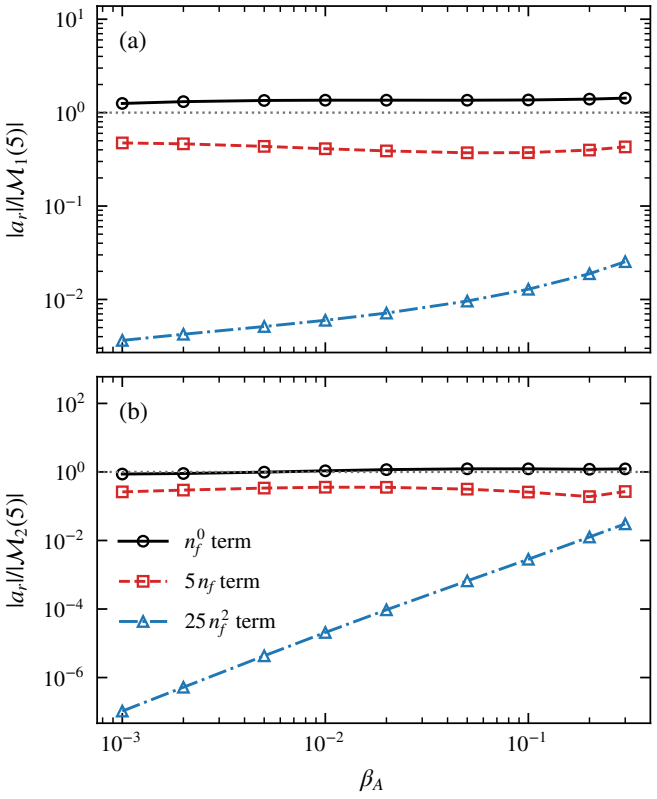


FIG. 6. Magnitudes of the three terms of Eq. (C1), divided by the magnitude of the complete coefficient, for $c = 1/3$, $m_A = \mu_R = 125\,\text{GeV}$ and $N_c = 3$. The horizontal line marks unity.

2×10^{-7} and b agrees with Eq. (B3) to 2×10^{-5}, the accuracy of the series at these velocities; the second derivative equals $6\beta_0^2\mathcal{M}^{(0)}$, the value implied by Eqs. (B1) and (B3), to 1×10^{-8}. Without the division by 64 the fit gives $a = 192$, and without the scale term of the one-loop remainders b differs from Eq. (B3) by $6\beta_0^2 \ln(m_A^2/\text{GeV}^2)$; the agreement therefore also fixes the reference scale of the published one-loop remainders at $\mu_R = 1$ GeV.

Origin of the factor 64. The Mathematica code of the original calculation documents it: the two-loop renormalisation step multiplies the O_G–O_G amplitude, including the tree and one-loop amplitudes that enter the infrared subtraction, by $64 = 8^2$, whereas the one-loop step does not. The pole cancellation of Ref. [7] is unaffected, since all terms of the two-loop subtraction carry the same factor. The released code and data use the normalisation of Eq. (23).

The original output. The Mathematica output of the original calculation confirms both corrections. Its one-loop remainders, normalised to the tree amplitude, depend on μ_R as in Eq. (B1) and agree with the first line of Eq. (23) to 5×10^{-56}; the published one-loop files coincide with them at $\mu_R = 1$ GeV. An earlier version of the two-loop renormalisation step, without the factor 64 and with the O_J contribution switched off, was evaluated at $x = 1/11$, $z = 3/11$. At that point the published P_1 remainder equals 64 times this result, multiplied by the tree amplitude, plus the O_G–O_J counterterm of the published version; the real parts agree to 5×10^{-31}. The imaginary parts differ by $32C_F$ times the one-loop O_G–O_J amplitude, a multiple of the tree amplitude. That earlier run is not an alternative result, however. Its renormalisation constant exceeds $Z_{GG}^2 = Z_{a_s}^2$ by $2\beta_1 a_s^2/\epsilon$, a pole with C_A^2 terms that no O_J contribution can supply, and it contains no O_G–O_J counterterm. With the same infrared subtraction as the published version, its single poles cannot cancel, and the difference is not an uncertainty of the published remainder. In the published version the normalisation of the O_G–O_J counterterm was fixed by the cancellation of the $C_F n_f$ single pole; a record of the original calculation from June 2019 shows that the same factor is required in both projections at seven kinematic points. The files and a script that reproduce these comparisons are part of the released material.

TABLE VIII. Evaluation time for the two complete Type II coefficients at the benchmark parameters, $c = 0$ and 80-digit working precision, for the unexpanded remainders (exact) and the series. First calls start with empty caches; repeated times are medians of seven evaluations at the same point. Starting the kernel and loading the files are excluded.

	first call [s]			repeated [s]		
β_A	exact	series	ratio	exact	series	ratio
1/10	56.04	11.61	4.83	0.3422	0.0178	19.18
1/5	45.68	11.64	3.92	0.3403	0.0179	18.99
3/10	44.64	11.64	3.84	0.3485	0.0181	19.22

Appendix C: Flavour decomposition

For $n_f = 5$ the complete coefficients are the sums

$$\mathcal{M}_i^{(2)}(5) = \mathcal{M}_{i,0}^{(2)} + 5\,\mathcal{M}_{i,1}^{(2)} + 5^2\,\mathcal{M}_{i,2}^{(2)} \equiv a_0 + a_1 + a_2. \tag{C1}$$

Figure 6 shows $|a_r|/|\mathcal{M}_i^{(2)}(5)|$. The n_f^0 term dominates both coefficients. The n_f^1 term is sizeable and partly cancels it: for $\mathcal{M}_1$ the quantity $1-|a_0+a_1+a_2|/(|a_0|+|a_1|+|a_2|)$ lies between 42.2% and 46.9% over the velocities shown. At leading power the number of flavours enters with the sign opposite to the gluonic terms, through $\beta_0 = (11N_c - 2n_f)/3$ in $\widehat{A}_{12}$, through $N_c a_\text{A} - n_f a_\text{F}$ in $\widehat{A}_{11}$ and through $4N_c - n_f$ in $\widehat{A}_{21}$ [Eqs. (39) and (40)], which accounts for the cancellation at small β_A. The n_f^2 term of $\mathcal{M}_2$ is suppressed as $\beta_A^2(1-c^2)$, Eq. (35).

Appendix D: Evaluation time and released data

The series through β_A^6 occupy 40 kB, against 204 MB for the published two-loop files, a factor of 5061. The evaluation time was measured before the reduction of Sec. III B, with the unexpanded expressions and the series in the forms used for the comparison of Sec. III C, at $\beta_A = 1/10$, $1/5$ and $3/10$, $c = 0$, $m_A = \mu_R = 125\,\text{GeV}$ and $N_c = 3$, with 80-digit working precision. On a first evaluation the series is 3.84–4.83 times faster. For repeated evaluations at the same point it is 18.99–19.22 times faster, with median 19.18; this does

not correspond to a scan over new points. The two timed representations occupy 0.61 MiB and 42.77 MiB on disk, a factor of 70.4. Table VIII gives the individual measurements. The measurements used Wolfram Language 13.2.0 for Linux x86 under Ubuntu 24.04.2 LTS on an AMD Ryzen 9 7950X processor with 124.9 GiB of memory.

The Supplemental Material [28], which is also attached to the arXiv version of this article as ancillary files, is a self-contained package. It contains the construction files, one for each flavour coefficient and power of β_A from β_A^{-4} to β_A^6; the series through β_A^6 in Wolfram Language form; the reductions of the endpoint constants; the 1080 validation records and the forward angle data; the tree-level projections; and an evaluator in Python and Wolfram Language. It also contains the Wolfram Language, C++ and Python code of the threshold construction and of the validation, the outputs of the original runs, against which the construction files and the validation records are checked, and a script that repeats the construction from the published expressions. The published expressions of Ref. [7] are downloaded and checked against checksums by a script. Each table, figure and number in this paper is generated from these data by a single command, including the check of the DPS weight in Fig. 3, the logarithms of Sec. V and the normalisation checks of App. B. A test suite checks the evaluator against the validation records. Only the construction of the series from the unexpanded expressions requires Mathematica. The series are intended for $\beta_A \leq 0.2$ at the angles tested (Sec. V C); outside this range the unexpanded remainders should be used.

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